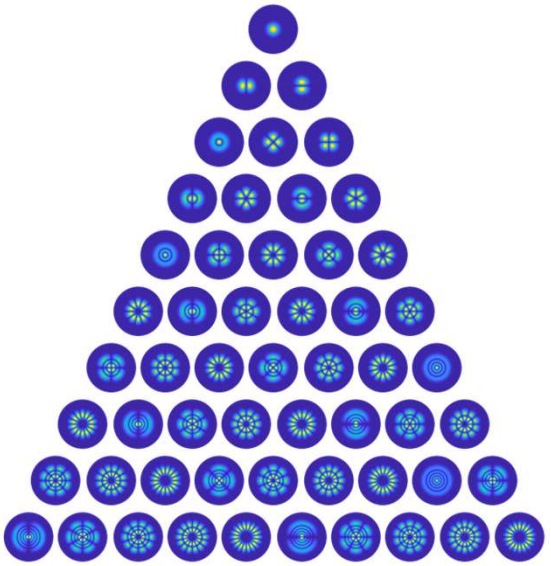
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

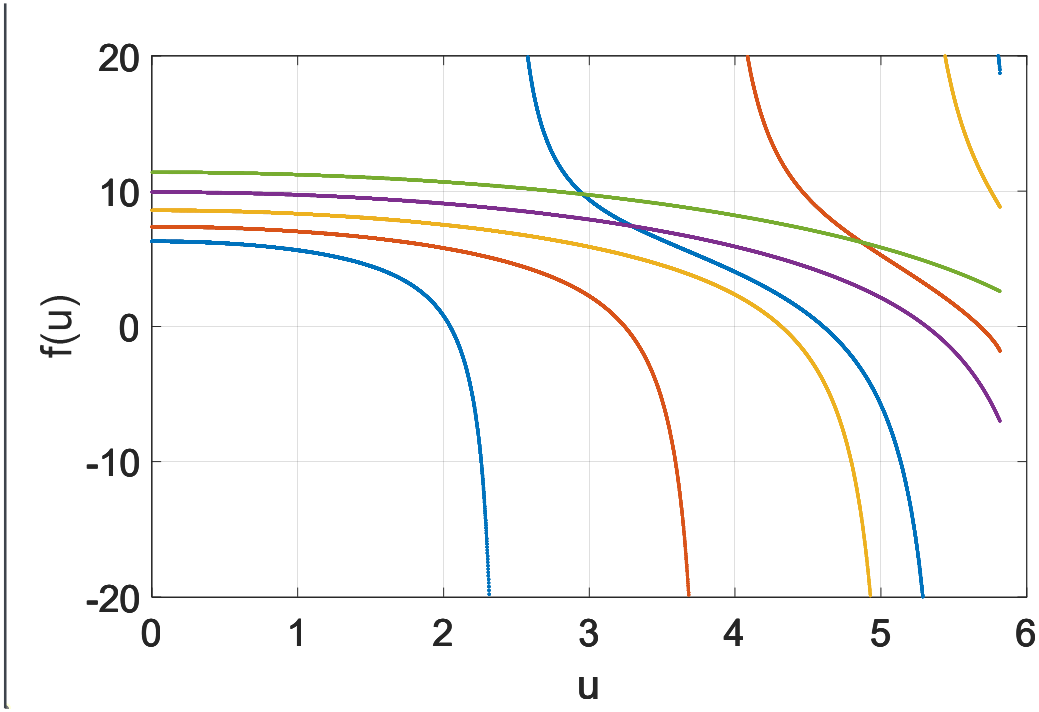
Lecture 6



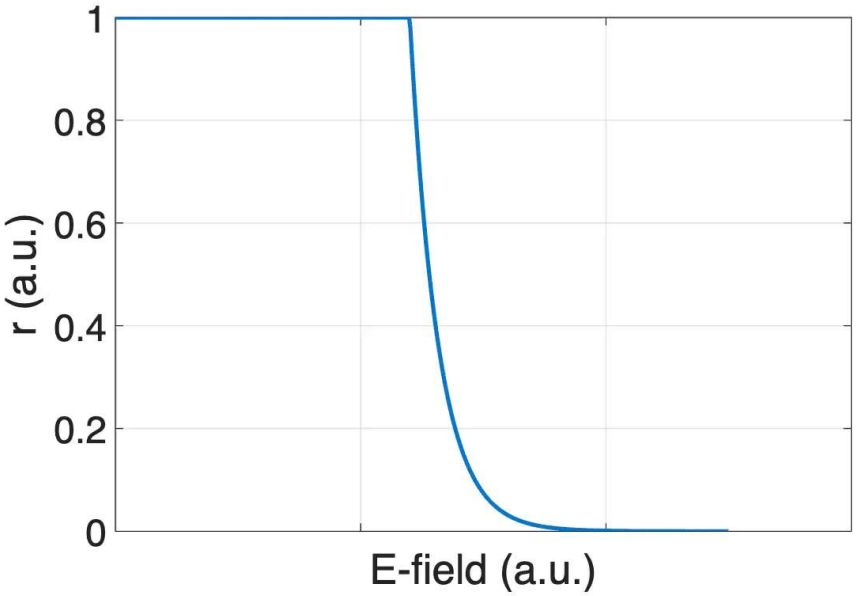
Prof. Dr.-Ing. Georg Rademacher

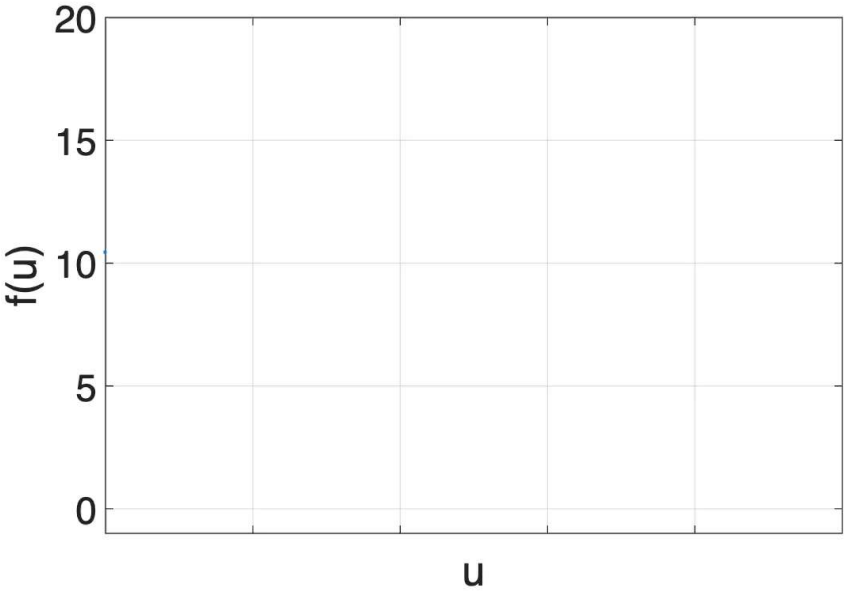
Linearly polarized (LP) waves

University of Stuttgart
Germany



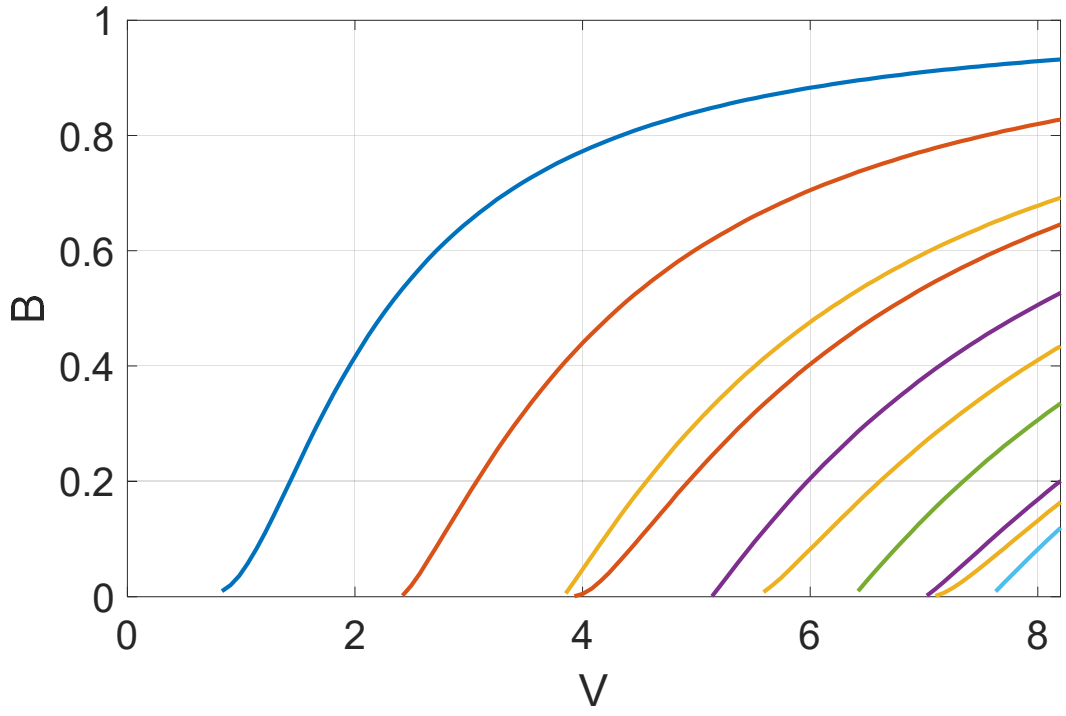
Linearly polarized (LP) waves





Linearly polarized (LP) waves

University of Stuttgart
Germany



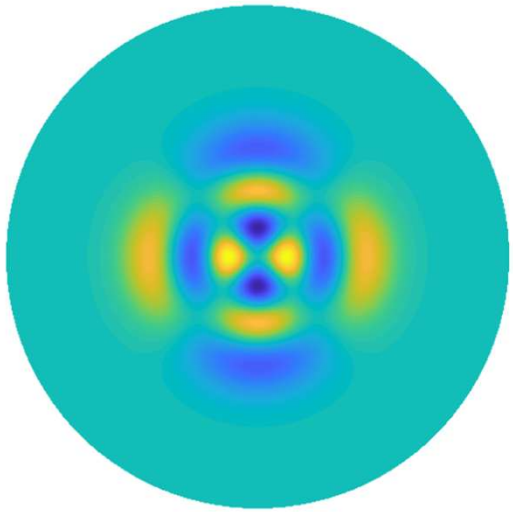


University of Stuttgart
Germany

Concept Test

University of Stuttgart
Germany

Which Linearly polarized (LP) mode is shown?



A) LP_{23}

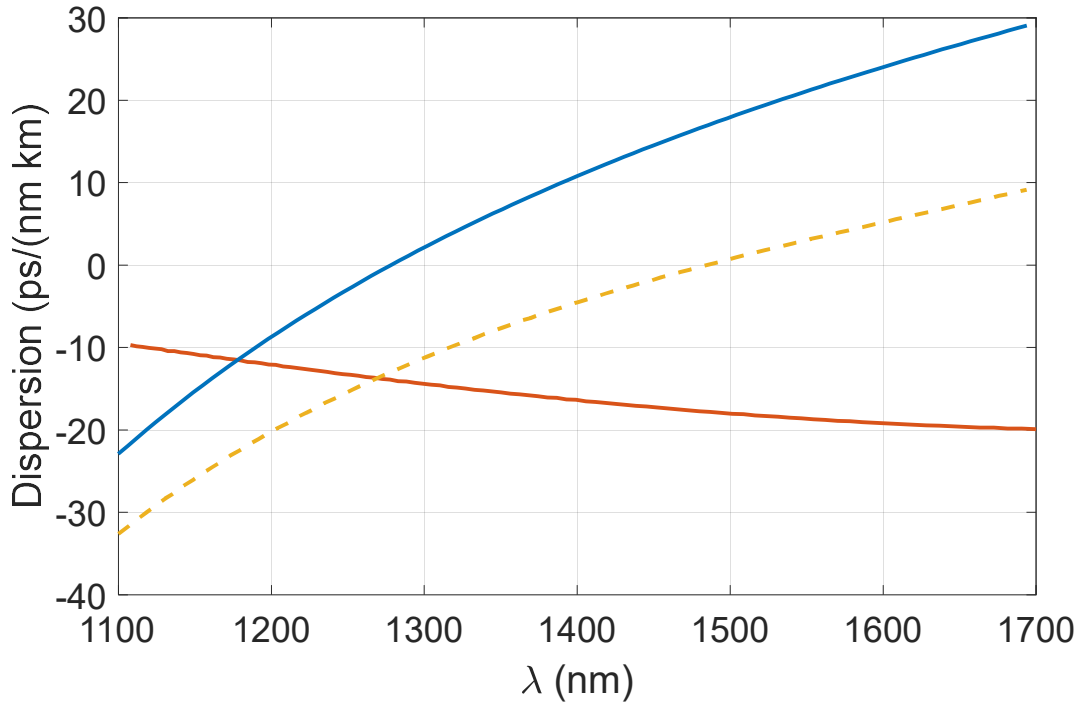
B) LP_{32}

C) LP_{42}

https://ilias3.uni-stuttgart.de/vote/K2XW

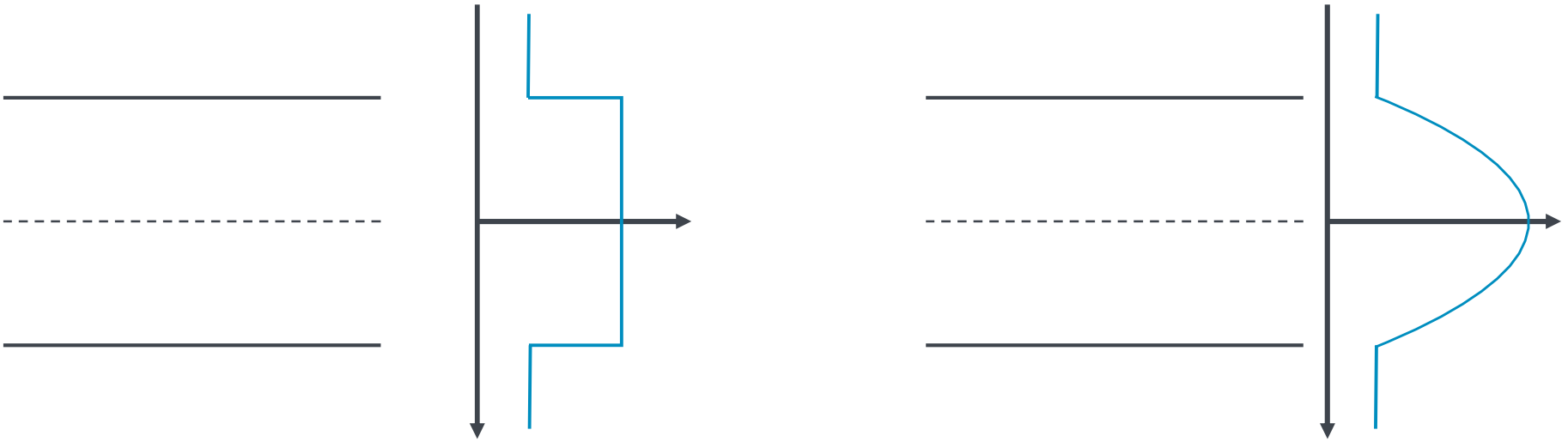
Chromatic dispersion of the fundamental mode

University of Stuttgart
Germany

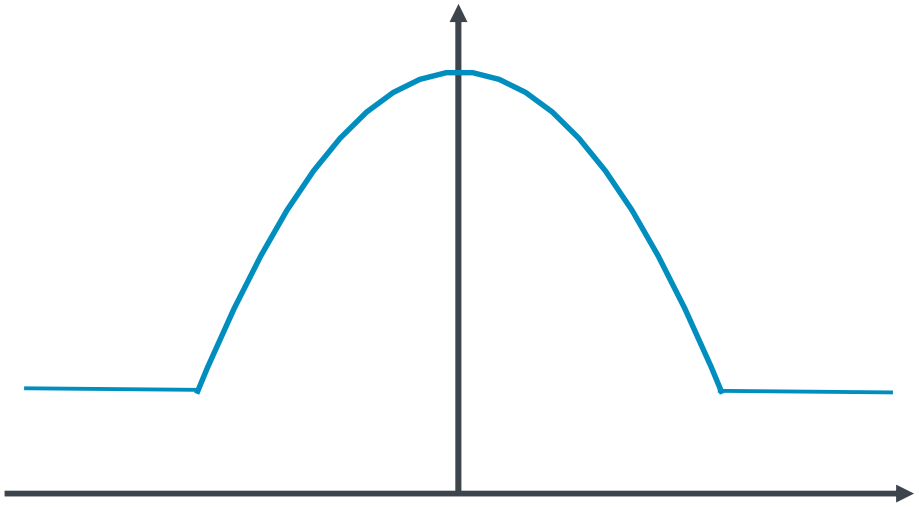


Graded-index fibers

University of Stuttgart
Germany



Graded-index fibers

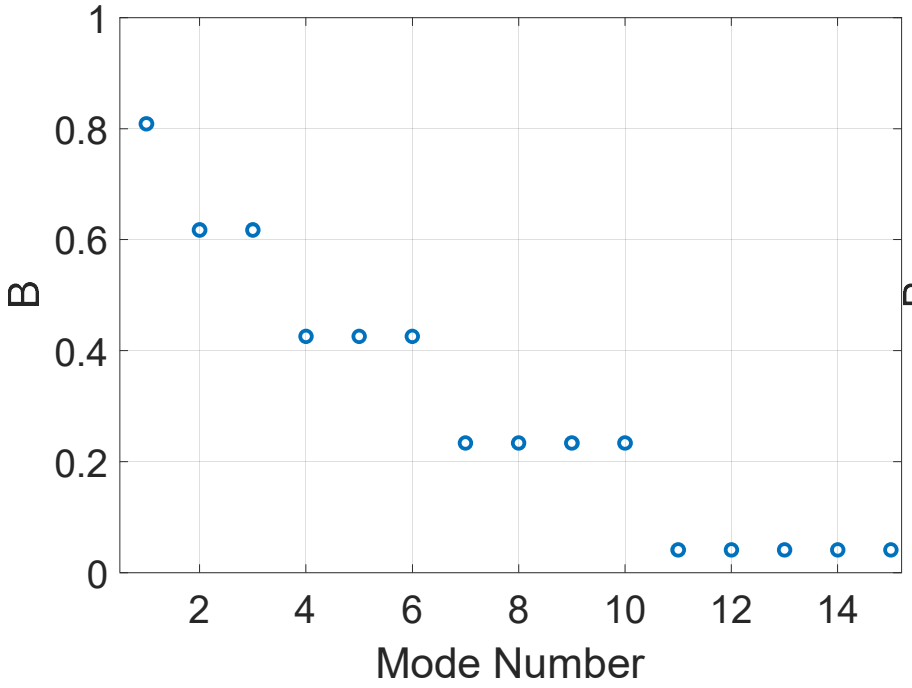


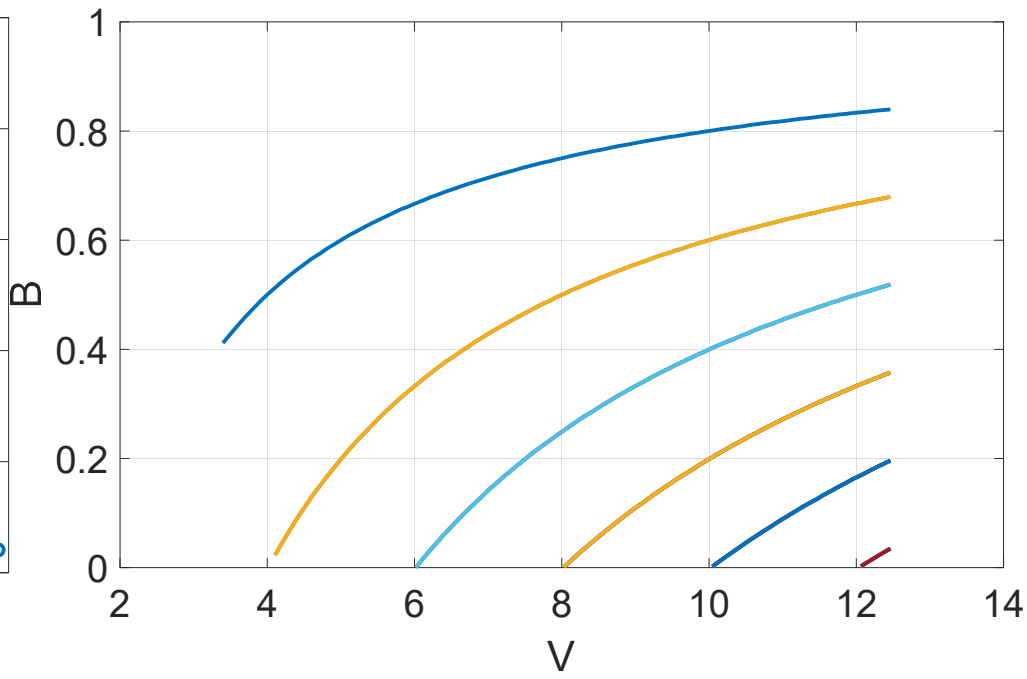
Graded-index fibers

Graded-index fibers

Graded-index fibers

University of Stuttgart
Germany







University of Stuttgart
Germany

Concept Test

Concept test

University of Stuttgart
Germany

For V = 9, which fiber supports more propagating modes?

A Graded index fiber

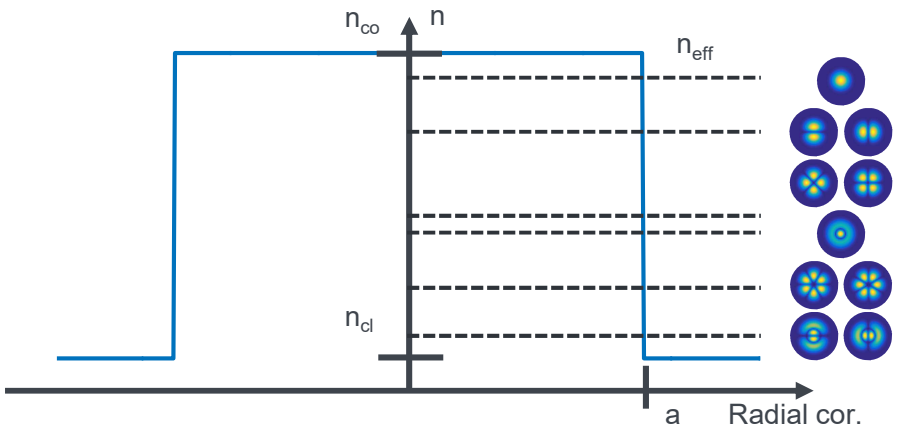
B Step index fiber

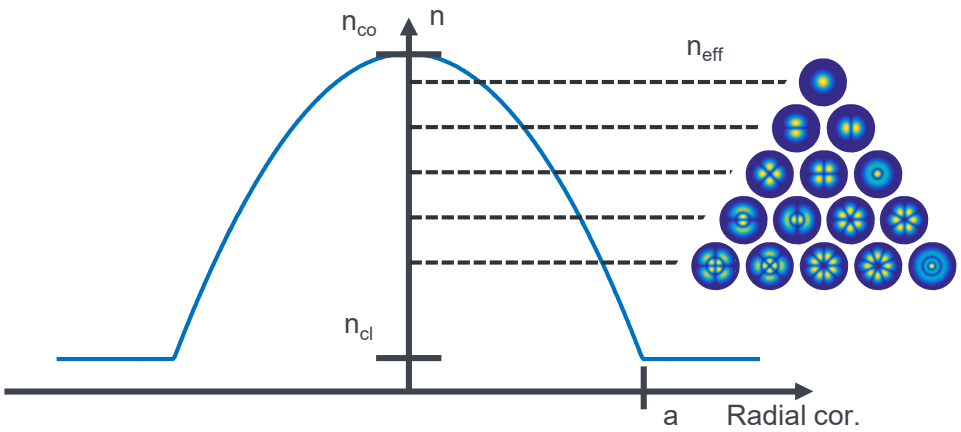
C Both have the same

https://ilias3.uni-stuttgart.de/vote/**VGN9**

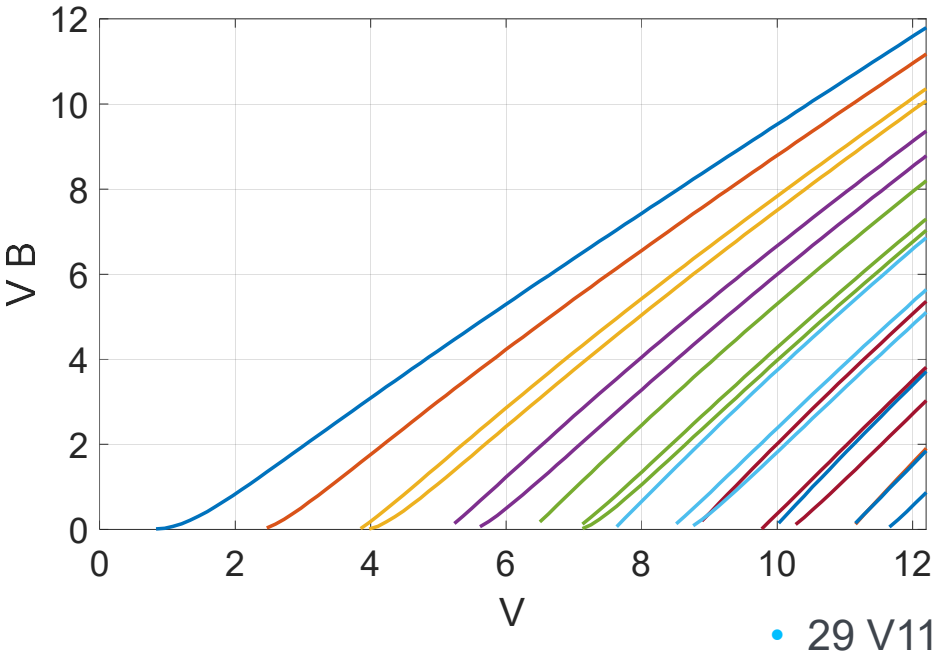
Multi-mode fibers (MMFs)

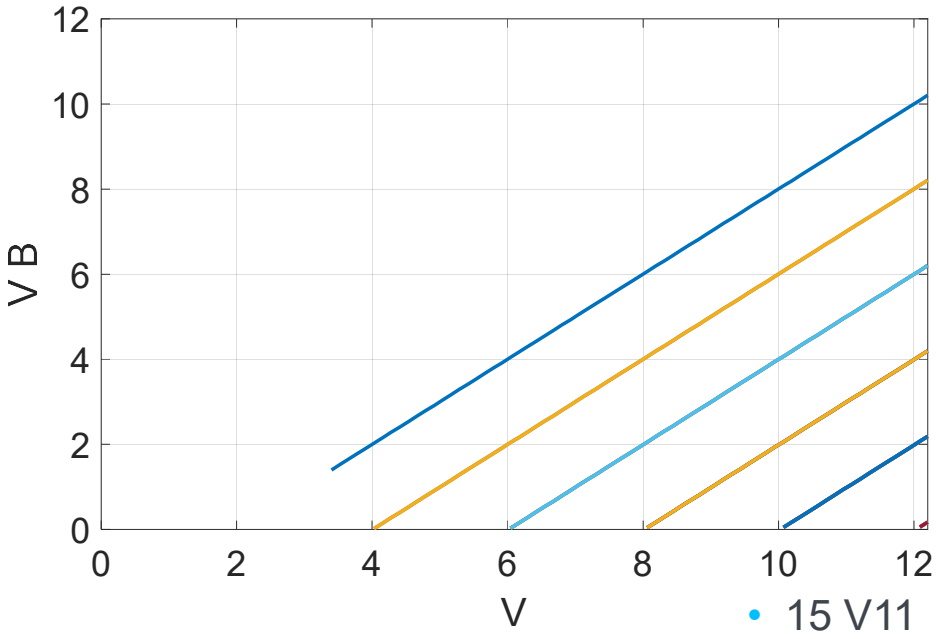
University of Stuttgart
Germany





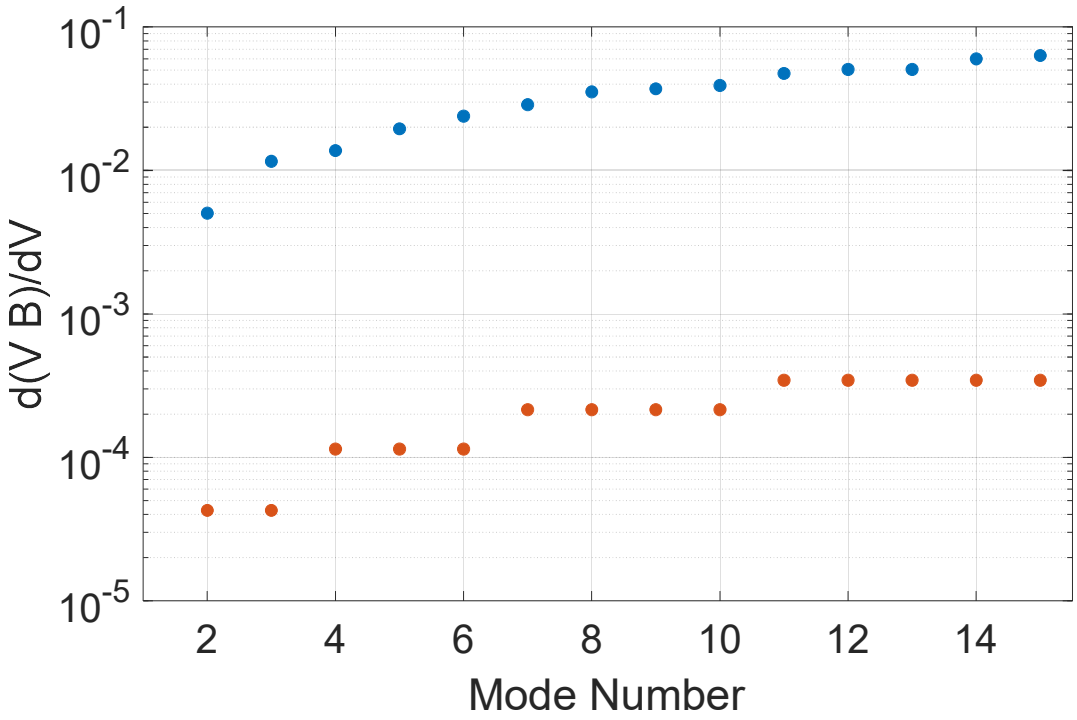
Graded-index fibers





Graded-index fibers

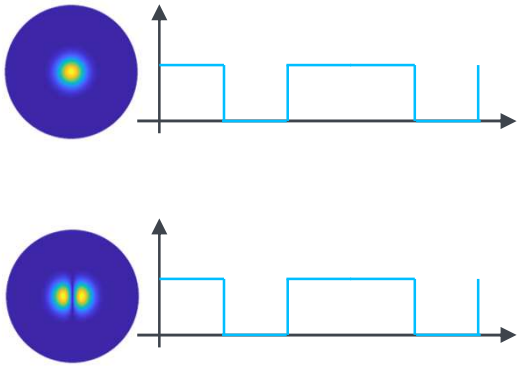
University of Stuttgart
Germany



Why is the delay difference so important?

University of Stuttgart
Germany





Why is the delay difference so important?

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Germany







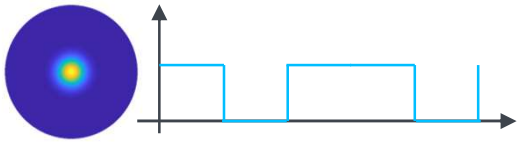


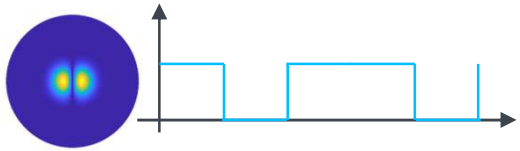


Why is the delay difference so important?

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Why is the delay difference so important?

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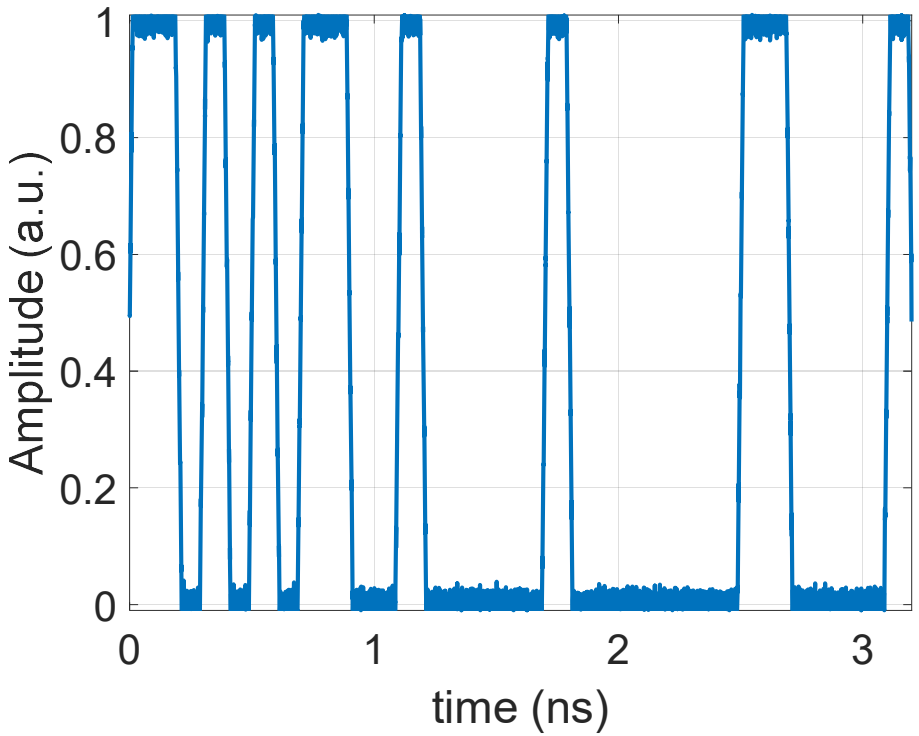


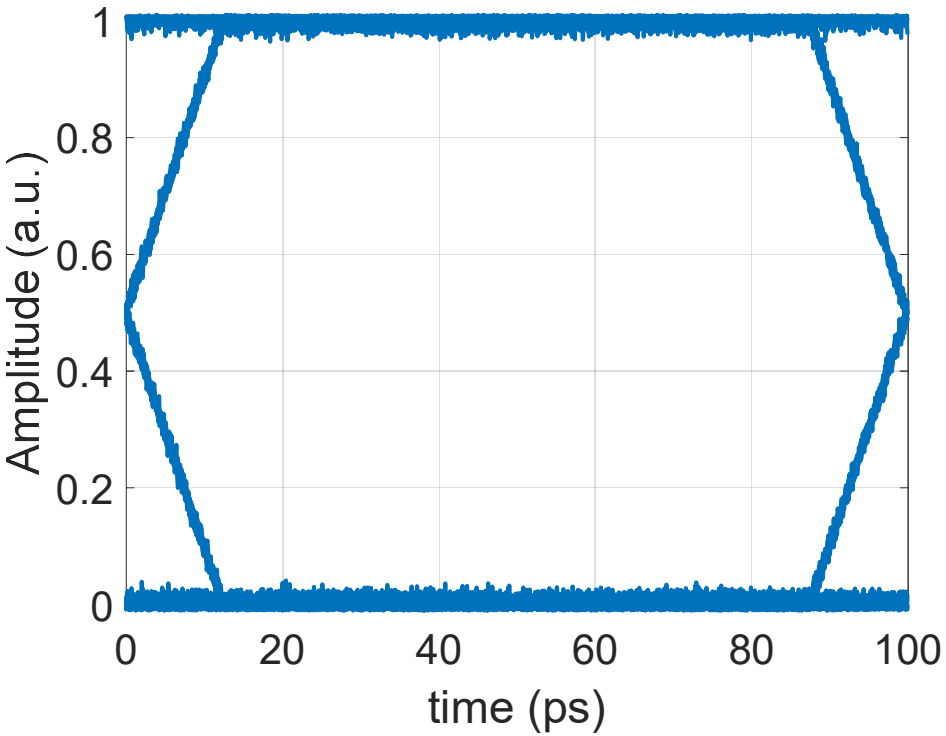




Eye-Diagram

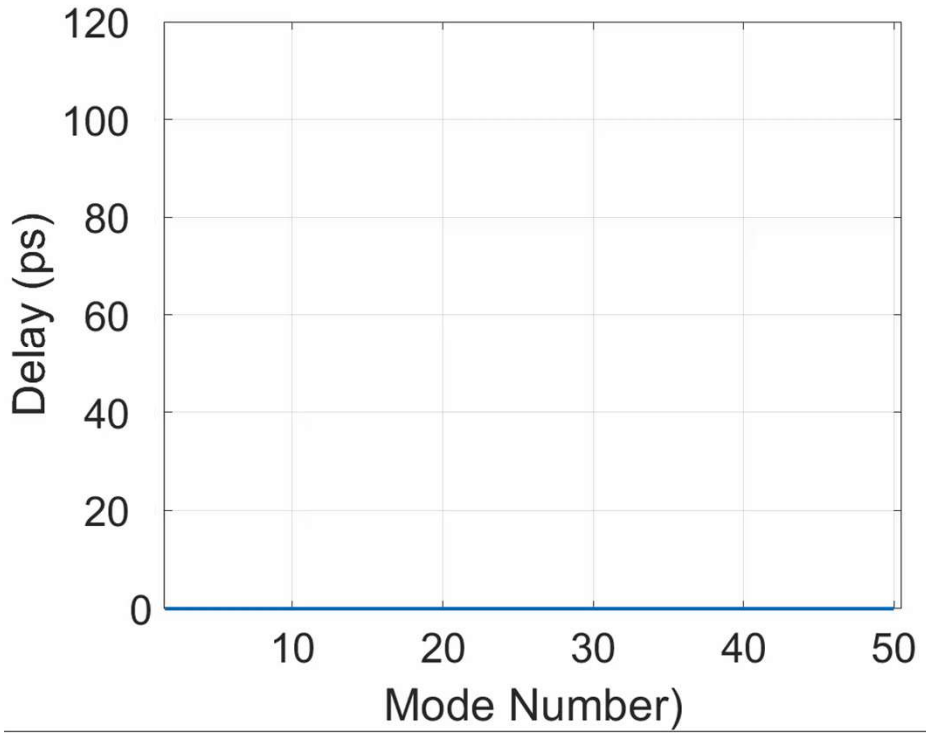
University of Stuttgart
Germany

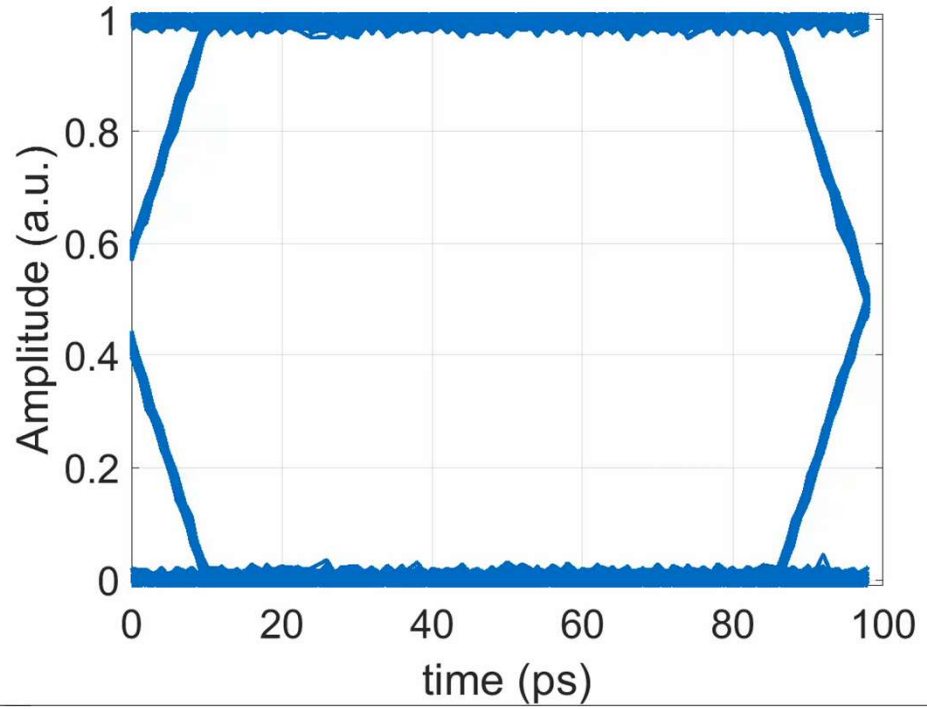




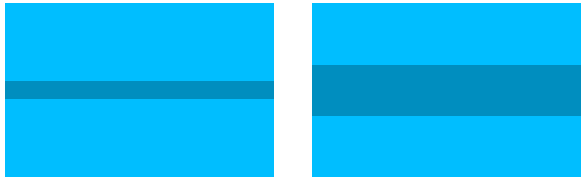
Why is the delay difference so important?

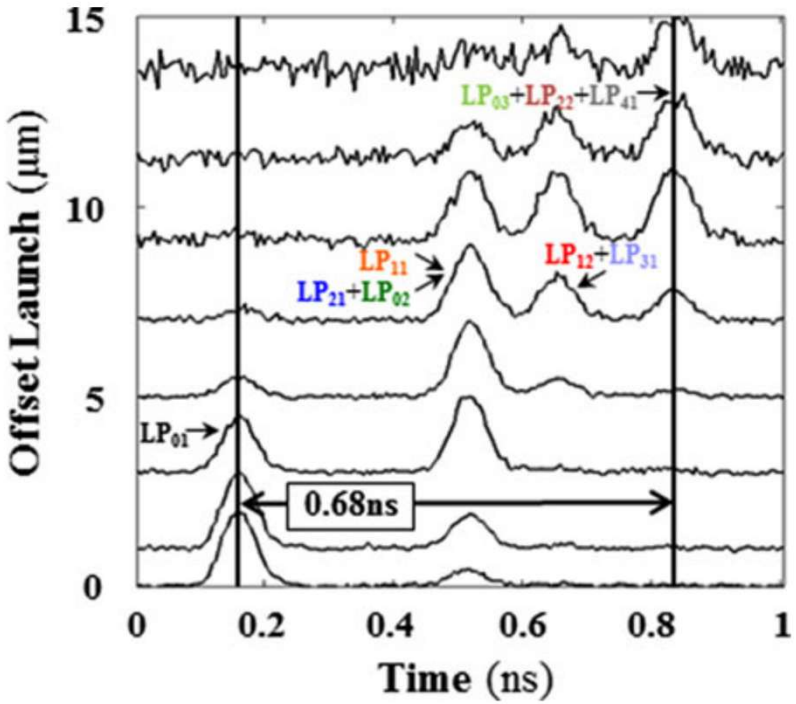
University of Stuttgart
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Graded-index fibers





P. Sillard et al., Low-differential-mode-group-delay 9-LP-mode fiber, JLT 2016



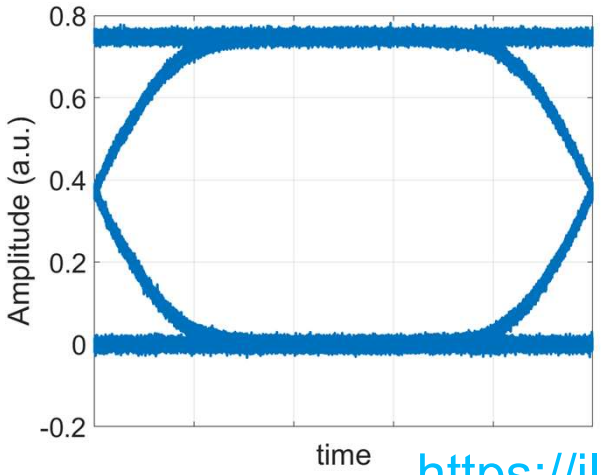
University of Stuttgart
Germany

Concept Test

Concept test

University of Stuttgart
Germany

How large can the differential mode-group-delay get so that the eye diagram is still open? Consider a Symbolrate of 50 Gbd, 10 km fiber length, and two modes.



A 20 ps/km

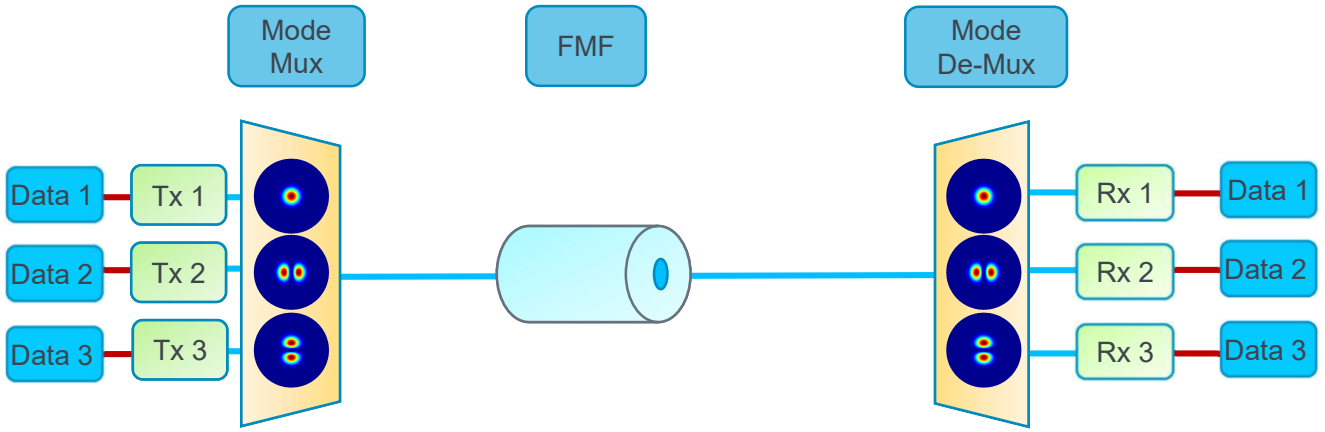
B 0.5 ps/km

C 1 ps/km

https://ilias3.uni-stuttgart.de/vote/**W4IQ**

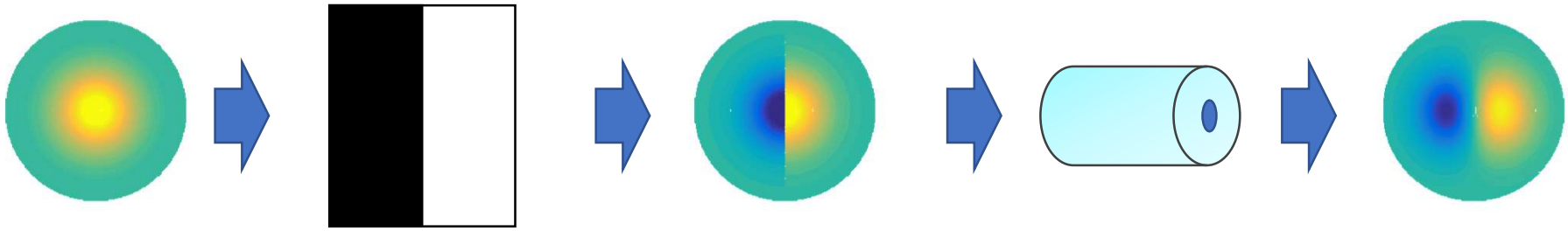
Space division multiplex with MMF

University of Stuttgart
Germany



Mode-Multiplexers

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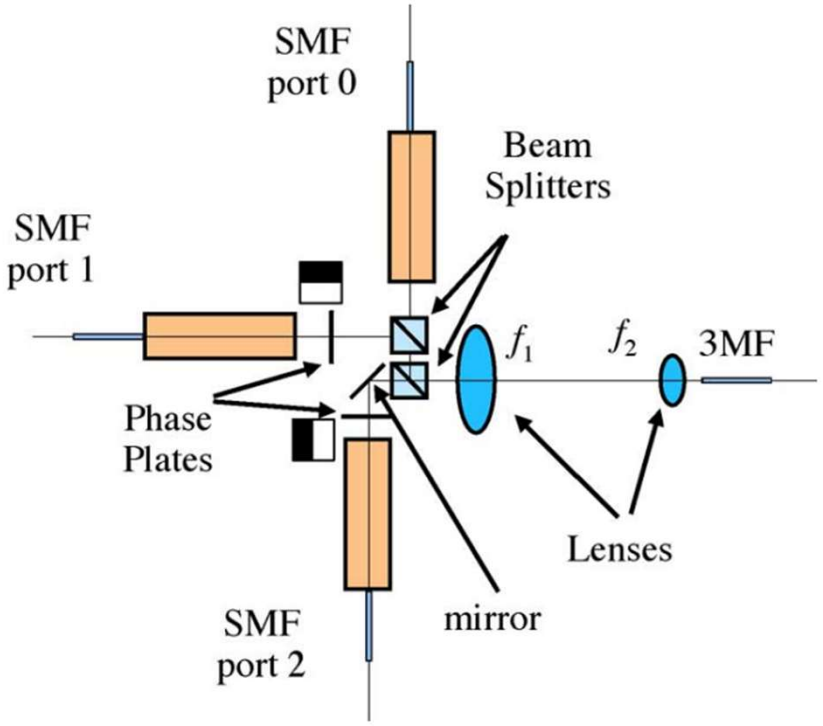
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 4, FEBRUARY 15, 2012

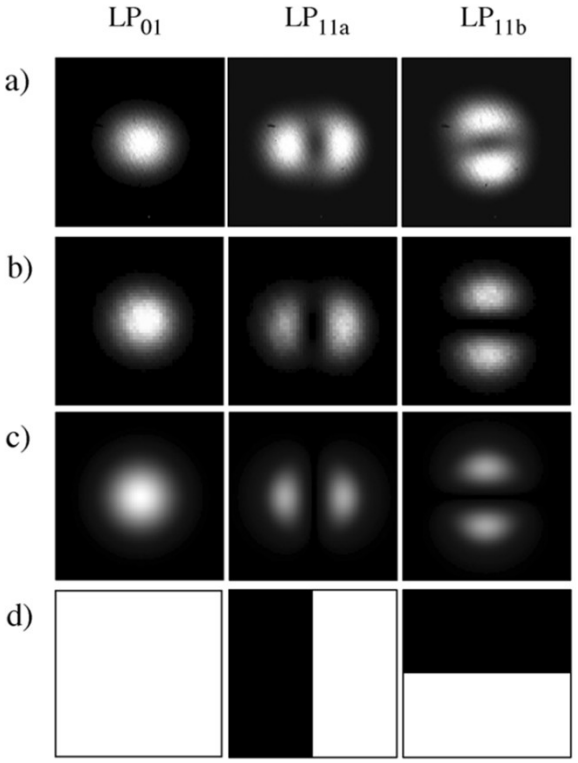
Mode-Division Multiplexing Over 96 km of Few-Mode Fiber Using Coherent 6×6 MIMO Processing

Roland Ryf, *Member, IEEE*, Sebastian Randel, *Senior Member, IEEE*, Alan H. Gnauck, *Fellow, IEEE, Fellow, OSA*, Cristian Bolle, Alberto Sierra, *Student Member, IEEE*, Sami Mumtaz, Mina Esmaeelpour, Ellsworth C. Burrows, René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Peter J. Winzer, *Fellow, IEEE, Member, OSA*, David W. Peckham, *Senior Member, IEEE*, Alan H. McCurdy, *Member, IEEE*, and Robert Lingle, Jr., *Member, IEEE*

(Invited Tutorial)

Mode-Multiplexers





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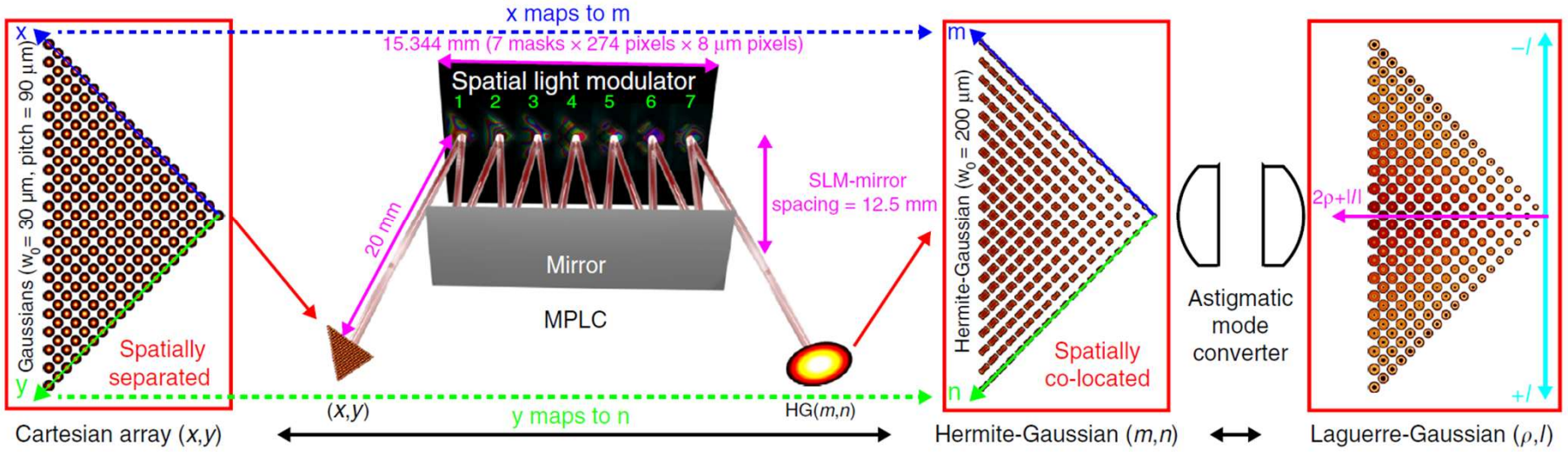
https://doi.org/10.1038/s41467-019-09840-4 OPEN

Laguerre-Gaussian mode sorter

Nicolas K. Fontaine[1,3], Roland Ryf[1], Haoshuo Chen[1], David T. Neilson[1], Kwangwoong Kim[1] & Joel Carpenter[2,3]

Multi-plane light conversion

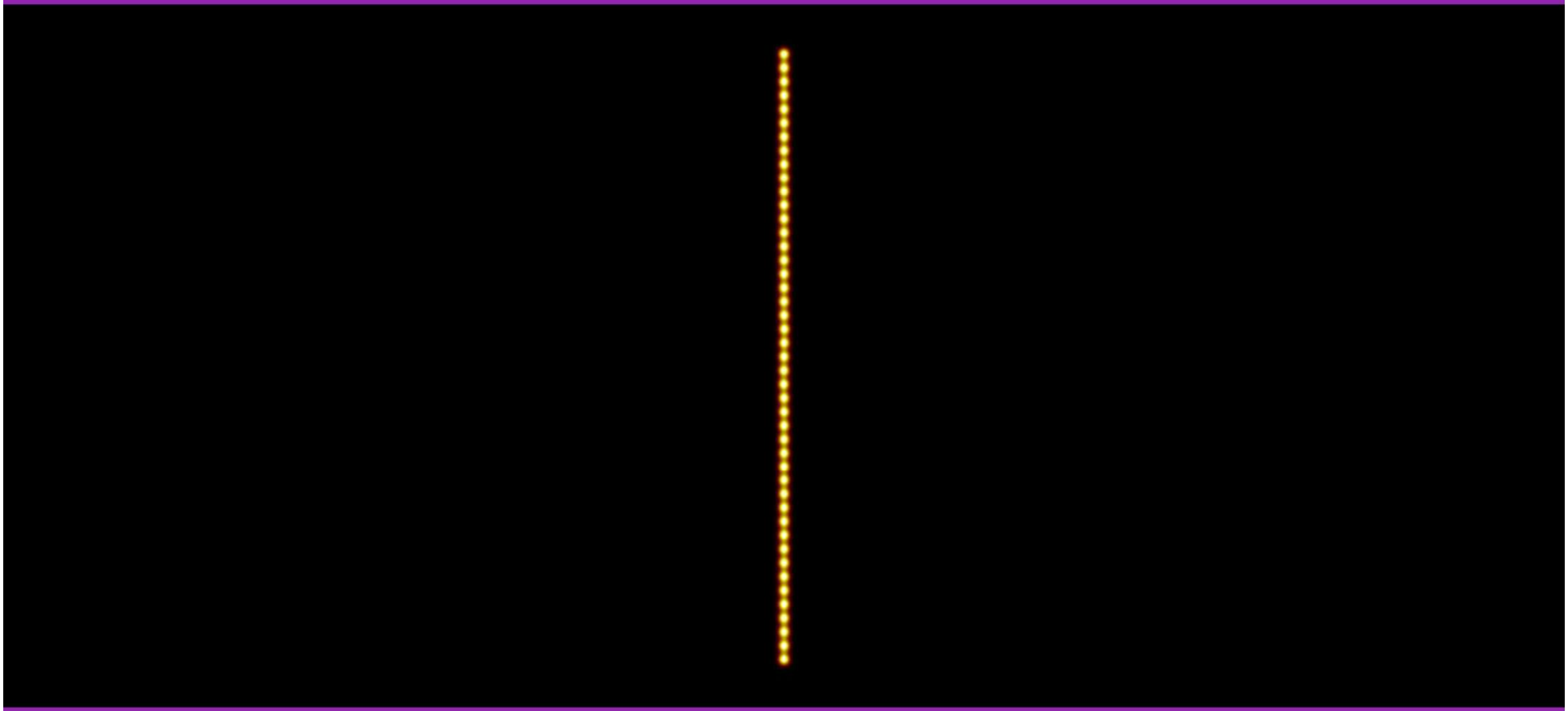
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Multi-plane light conversion

NOKIA Bell Labs

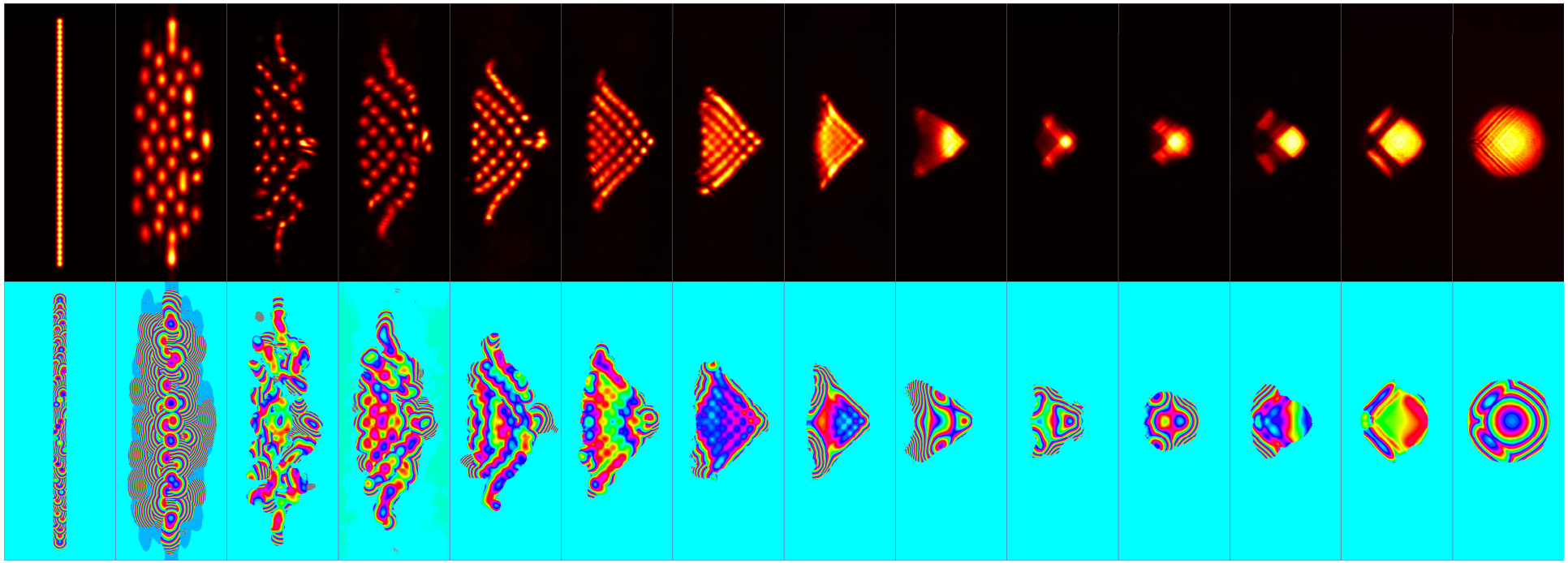


J.-F Morizur et al., J. Opt. Soc. Am. A, 27 (11) 2524-2531 (2010).

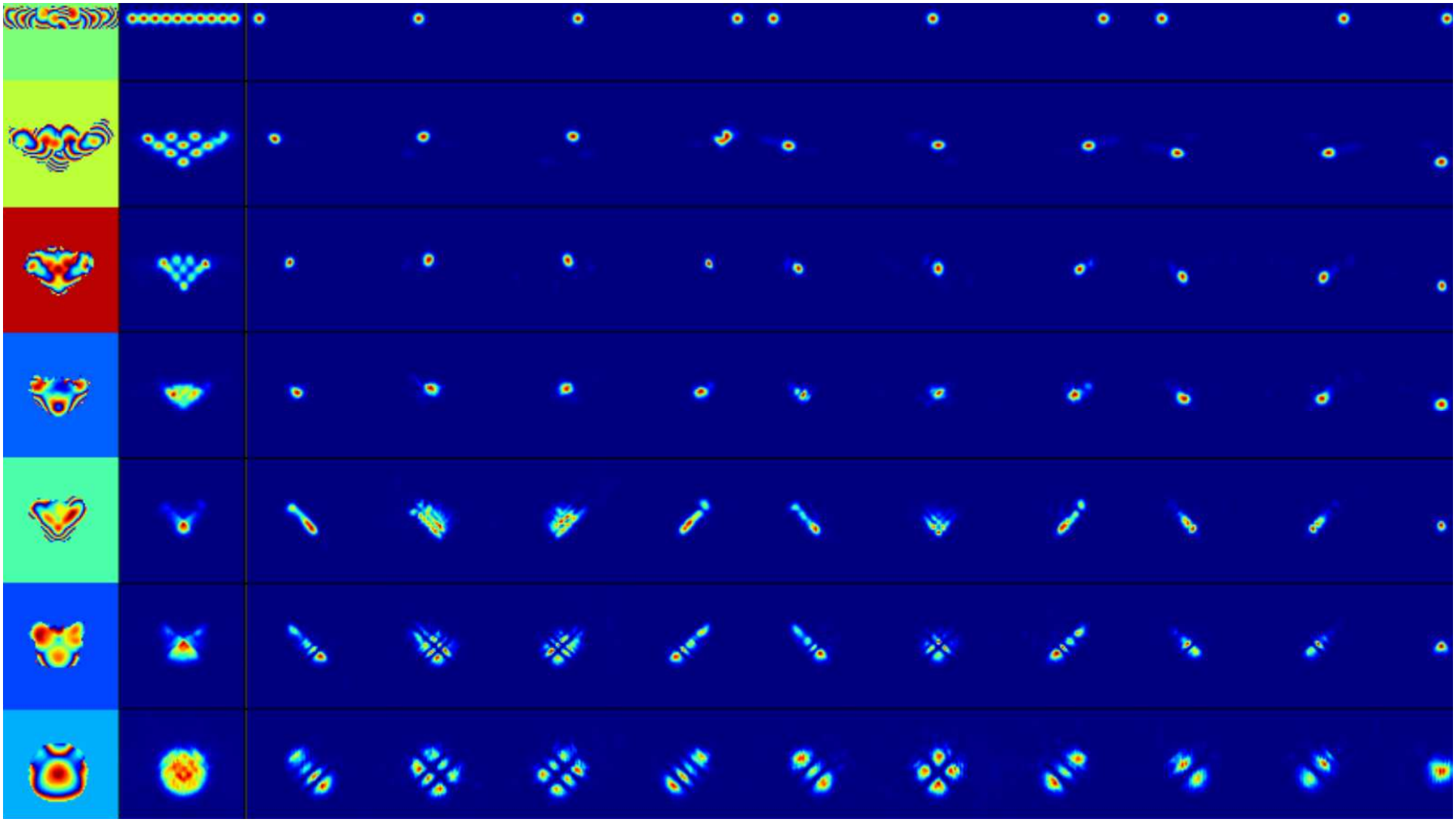
THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Triangular → HG

NOKIA Bell Labs

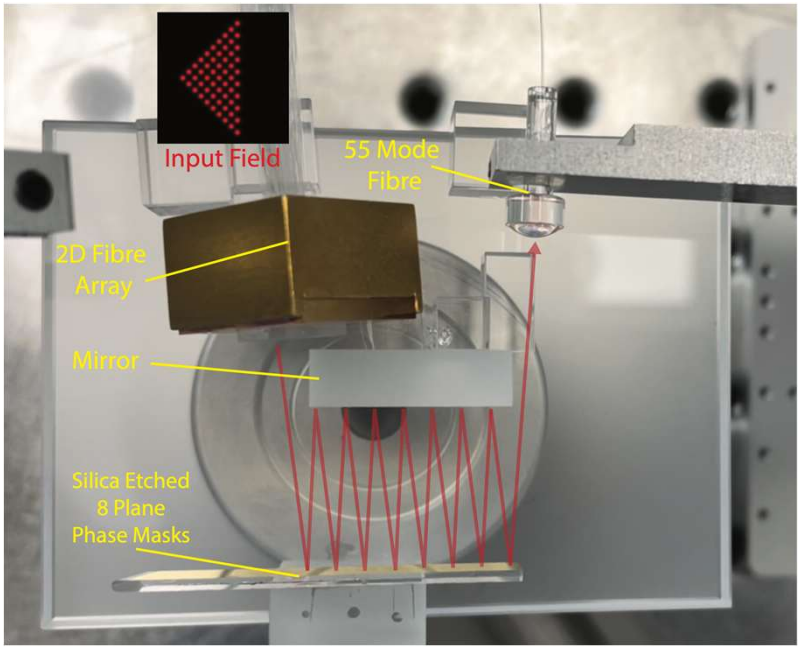


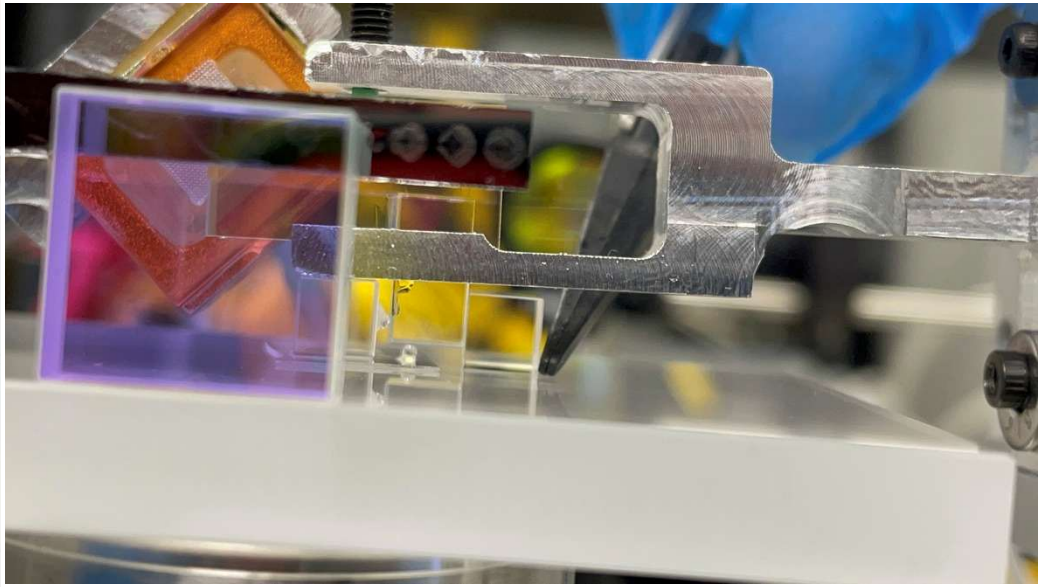
45 modes, 14 planes (512×1280 5μm pixels @ 1550nm, 24mm plane spacing)



Mode-Multiplexers

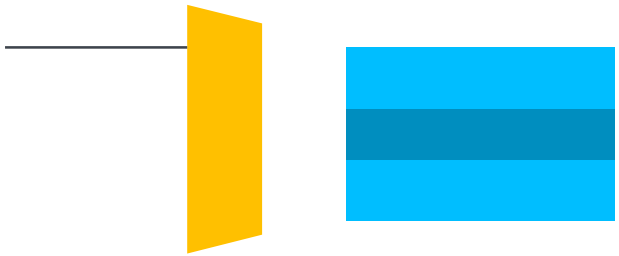
University of Stuttgart
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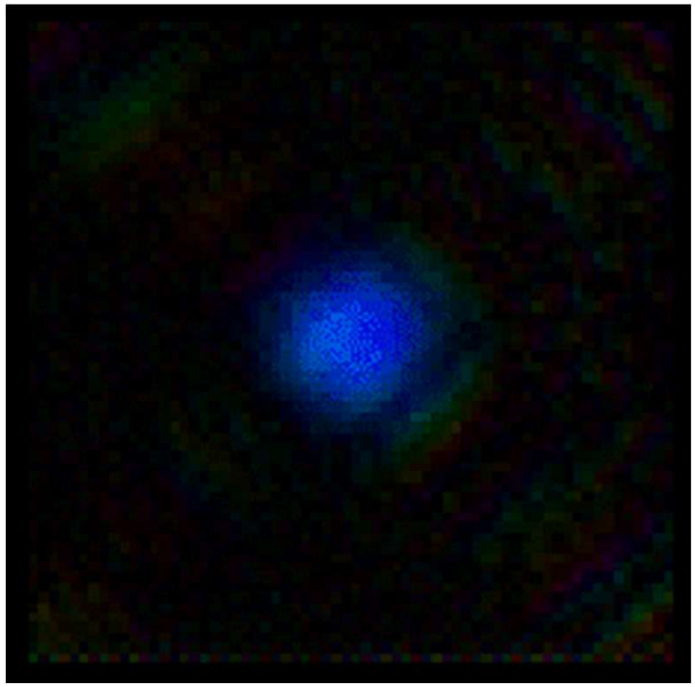




Mode-Multiplexers

University of Stuttgart
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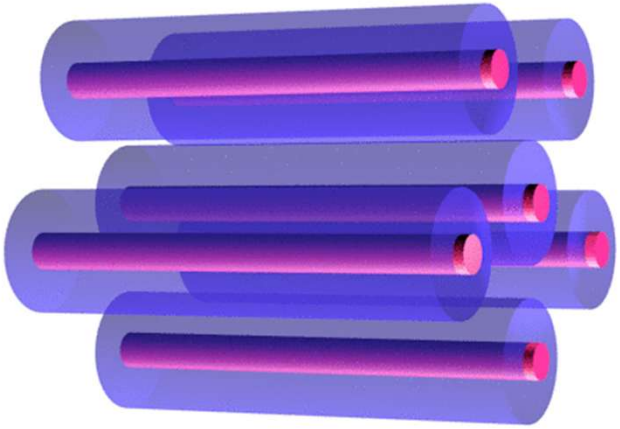


Photonic Lanterns, 3-D Waveguides, Multiplane Light Conversion, and Other Components That Enable Space-Division Multiplexing

This article reviews the mode-shaping technologies needed to perform mode-division multiplexing on few-mode fibers and discusses implementations and tradeoffs of four different techniques: multiplane light conversion, fused fiber devices, 3-D waveguides in glass, and free-space imaging systems.

By NICOLAS K. FONTAINE, *Senior Member IEEE*, JOEL CARPENTER, *Member IEEE*, SIMON GROSS, SERGIO LEON-SAVAL, YONGMIN JUNG, DAVID J. RICHARDSON, *Fellow IEEE*, AND RODRIGO AMEZCUA-CORREA

Mode-Multiplexers: Photonic Lantern



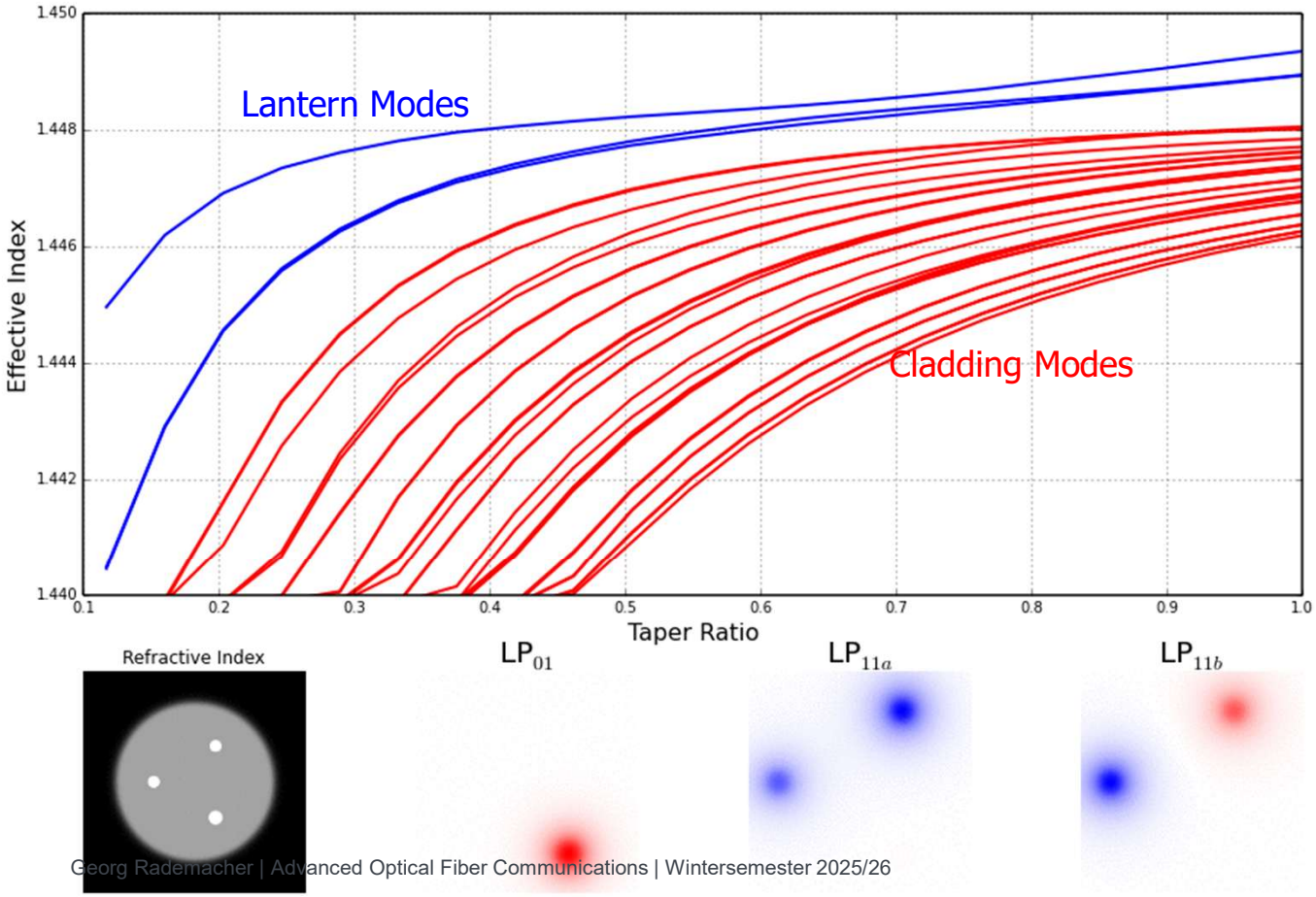


issimilar Fiber Mode Selective Lantern
diabatic transition from groups of cores to modes.

iversity of Stuttgart
rmany

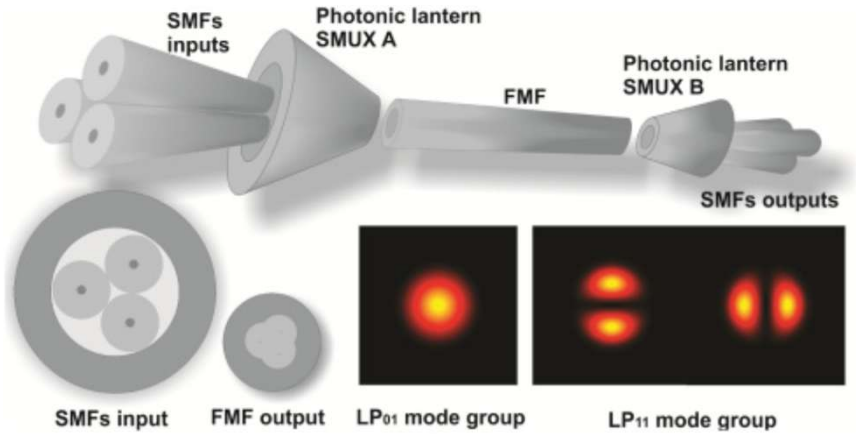
01 Big Core

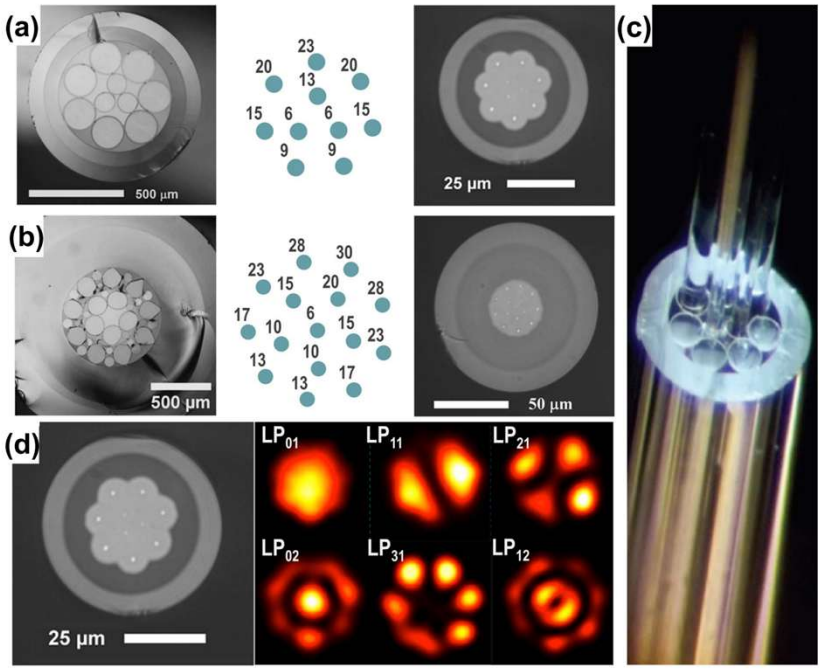
11 Two Smaller Cores

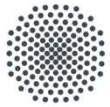


Mode-Multiplexers

University of Stuttgart
Germany







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Germany

Nanophotonics 2015; 4:332–352

DE GRUYTER OPEN

Review Article

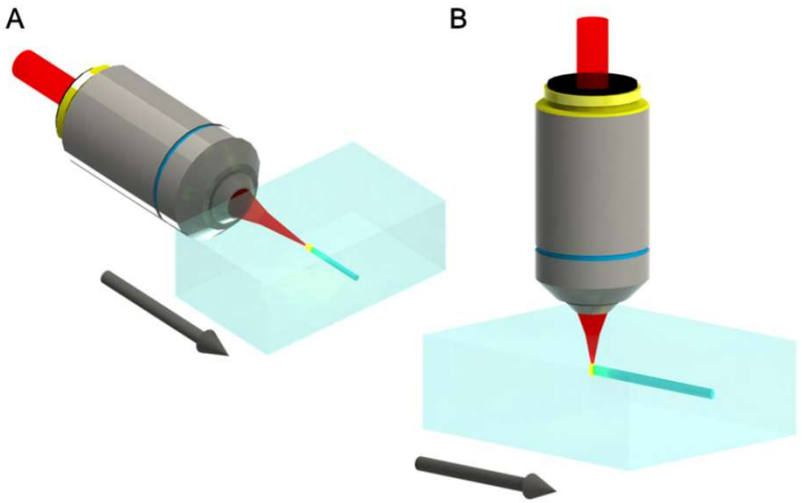
Open Access

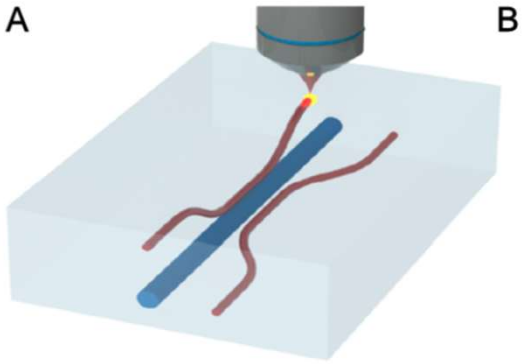
S. Gross* and M. J. Withford

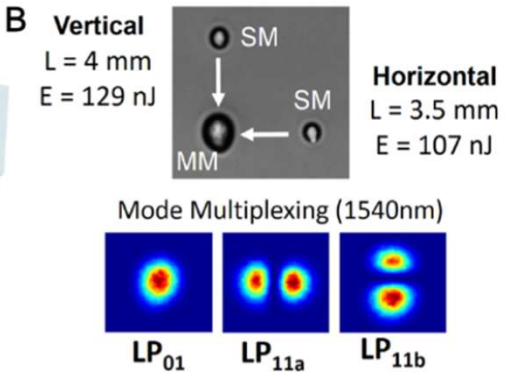
Ultrafast-laser-inscribed 3D integrated photonics: challenges and emerging applications

Mode-Multiplexers

University of Stuttgart
Germany



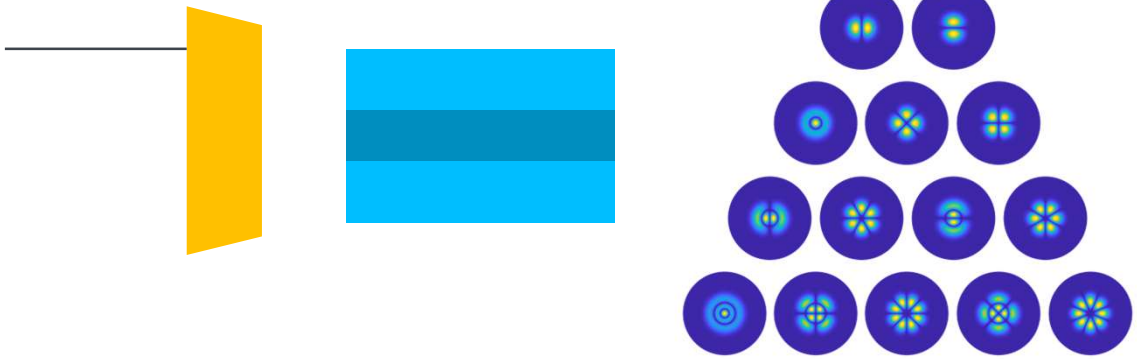


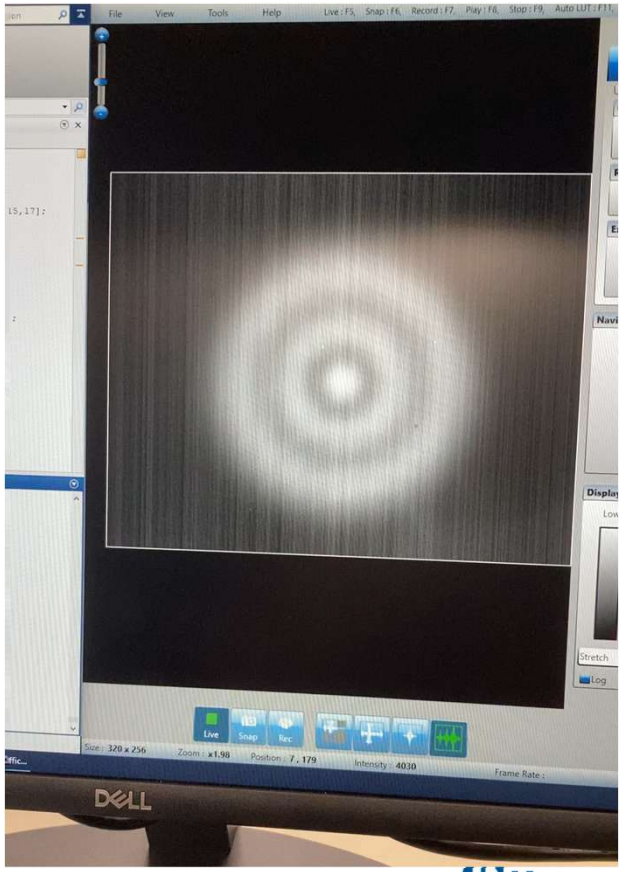


Graded-index fibers

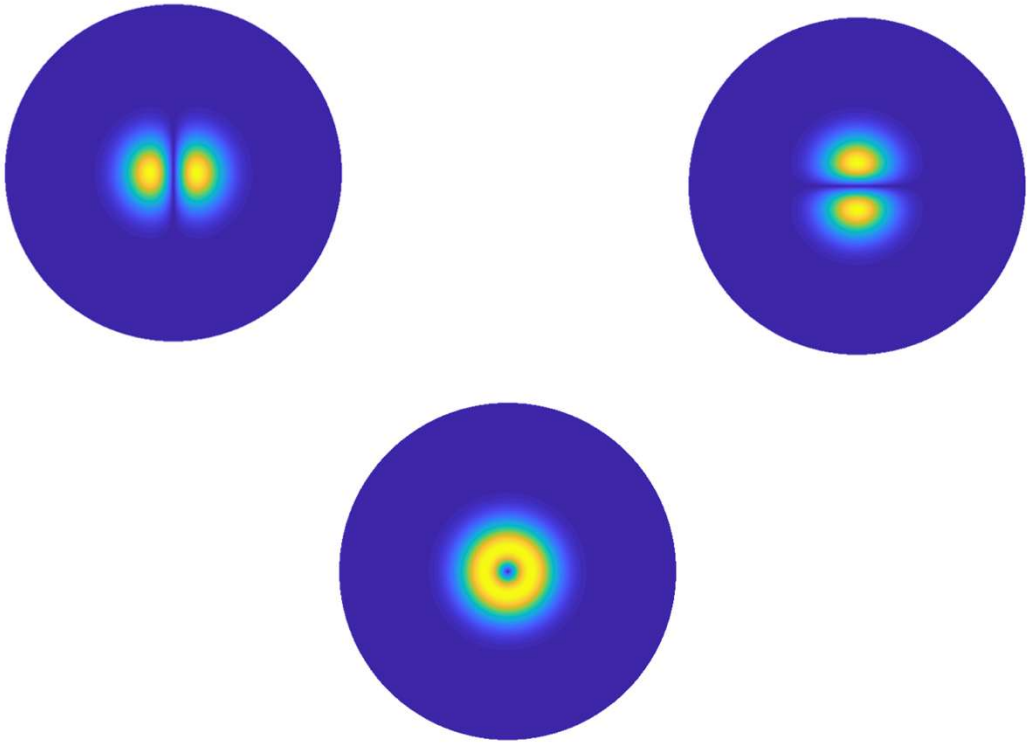
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Mode Coupling



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Thank you!



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